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Lab Exercise 02 Report



FitFlow Redesign

Data Analysis, Personas & Requirements – Lab Exercise 02 Report

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Case Study	FitFlow Redesign Scenario

Contents

Case Study Overview – FitFlow Redesign	3
Activity 1: Organize Interview / Survey / Observation Data.....	4
Activity 2: Perform Thematic Analysis	6
Activity 3: Create an Affinity Diagram.....	7
Activity 4: Develop Three Personas and Empathy Maps	9
Persona 1: Alex Rivera – Busy Professional (32).....	9
Persona 2: Priya Singh – Beginner Teacher (27).....	10
Persona 3: Sam – Fitness Enthusiast (29)	12
Activity 5: Write User Stories	15
Activity 6: Create User Journey Maps	16
Current-State Journey (Alex)	16
Future / Ideal-State Journey (Alex).....	16
Activity 7: Generate How Might We Statements	20
Activity 8: Functional & Non-Functional Requirements with Traceability	21
Functional Requirements.....	21
Non-Functional Requirements	21
Traceability Matrix.....	22

Case Study Overview – FitFlow Redesign

FitFlow is a fitness-tracking app from a mid-sized health-tech startup that had been steadily losing users over the past year. App-store ratings fell from an average of 4.6 to 3.8 stars and user retention dropped sharply, with 68% of new users abandoning the app shortly after onboarding. In Lab Exercise 01, user research was planned and conducted with primary and secondary stakeholders to uncover the reasons behind these problems. This report (Lab Exercise 02) takes the raw data gathered in that research and analyses it – organising the data, performing thematic analysis, building an affinity diagram, creating personas and empathy maps, writing user stories and journey maps, generating How-Might-We statements, and finally producing a prioritised requirements list with a traceability matrix to guide the redesign.

Key Figures Carried Forward from Lab 01

Indicator	Before Redesign	Target / After Redesign
Average app-store rating	3.8 stars (down from 4.6)	Recover to ~4.5 stars
Users abandoning after onboarding	68%	Significantly reduced
User retention (past year)	Declining	Improve by ~35%
Usability score (SUS)	68 / 100	87 / 100

The three redesign priorities that emerged from Lab 01 – AI-powered personalised workouts, social community features, and improved nutrition tracking – drive the analysis throughout this report.

Activity 1: Organize Interview / Survey / Observation Data

All raw research data from Lab 01 was gathered, cleaned, and organised in a central location before analysis. The sources included interview transcripts, survey responses, usability-testing notes, app-analytics reports (Firebase and Mixpanel), and app-store reviews.

Data Sources Gathered

Source	Contents	Volume (approx.)
Interview transcripts	Semi-structured interviews with users and trainers	5 sessions
Survey responses	15-question questionnaire (Typeform / in-app)	60+ responses
Observation / usability notes	Contextual-inquiry notes during workout & logging	3 sessions
App-analytics reports	Retention, onboarding drop-off, feature usage	Firebase, Mixpanel
App-store reviews	Public reviews and star ratings	50+ reviews

Data-Cleaning Steps

- Removed duplicate survey entries and blank or incomplete responses.
- Excluded off-topic app-store reviews (e.g., billing complaints unrelated to usability).
- Standardised wording and spelling so similar comments could be grouped later.
- Anonymised all records using participant IDs (P01–P05) before analysis.

Categorised Data (Central Spreadsheet Sample)

The cleaned data was stored in a central spreadsheet and tagged by source, demographic, and initial topic. A sample of the organised records is shown below.

ID	Source	Topic Tag	Notable Quote / Observation
P01	Interview	Motivation	“Finds the current workout plans repetitive; wants challenges with friends.”
P02	Interview	Workout experience	“Cannot follow fixed plans – needs workouts that fit a busy schedule.”
P03	Observation	Onboarding	Stalled during setup; never reached a first workout.

ID	Source	Topic Tag	Notable Quote / Observation
P05	Interview	Nutrition tracking	“Manual food logging is time-consuming; users stop after a few days.”
P06	App reviews	Personalisation	Multiple reviews call the plans “generic” and “not tailored to me.”
P07	Analytics	Onboarding	68% of new users drop off shortly after onboarding.

Activity 2: Perform Thematic Analysis

The organised data was reviewed several times and key quotes and observations were coded. Related codes were then grouped into major themes and sub-themes, and the frequency and strength of each theme were recorded with supporting evidence.

Coding and Themes

Theme	Sub-themes / Codes	Frequency	Supporting Evidence
Lack of personalisation	Generic plans; no adaptation; plans ignore goals	Very high	P01, P02, P04; app reviews call plans “generic”.
Poor motivation	Repetitive routines; no rewards; no accountability	High	P01, P04; users want challenges and achievements.
Nutrition-tracking issues	Tedious manual logging; abandoned after days	High	P05; survey shows low nutrition-tracking frequency.
Onboarding friction	Confusing setup; early drop-off	High	P03 observation; analytics 68% drop-off.
Social isolation	Exercising alone; wants community	Medium	P02, P03, P05; requests for groups and challenges.
Need for flexibility	No time; wants short adaptive workouts	Medium	P02; “quick 15-minute workout options.”

Theme Insights

- Lack of personalisation is the strongest and most frequently mentioned theme – it appears across interviews, observations, and app reviews.
- Poor motivation and social isolation are closely linked: users stay engaged when the app provides challenges, rewards, and a sense of community.
- Nutrition-tracking issues and onboarding friction are practical usability blockers that cause early and ongoing drop-off.
- Need for flexibility cuts across personas – busy users need workouts that adapt to the time they actually have.

The themes were validated by cross-checking each one against at least two independent data sources (for example, an interview quote plus an analytics figure or app review), confirming they are accurate and complete.

Activity 3: Create an Affinity Diagram

Individual insights, user quotes, and observations were written on digital sticky notes and sorted into natural clusters on a digital whiteboard (Miro). Each cluster was given a clear theme name, and relationships between clusters were identified. The structure of the affinity diagram is summarised below.

Affinity Clusters

Cluster (Theme)	Grouped Sticky Notes (Insights / Quotes)	Related Cluster
Personalisation	“Plans feel generic” · “Not tailored to me” · “Want AI recommendations” · “Adapt to my goals”	Flexibility
Motivation & Rewards	“Routines are repetitive” · “Want achievement badges” · “Challenges with friends”	Social & Community
Social & Community	“Exercising alone is hard” · “Want groups” · “Accountability partners”	Motivation & Rewards
Nutrition Tracking	“Logging is tedious” · “Stop after a few days” · “Want camera-based logging”	Onboarding & Usability
Onboarding & Usability	“Got lost in setup” · “Didn’t reach first workout” · 68% drop-off	Personalisation
Flexibility	“No time” · “Quick 15-min workouts” · “Smart reminders”	Personalisation

Key Findings from the Affinity Diagram

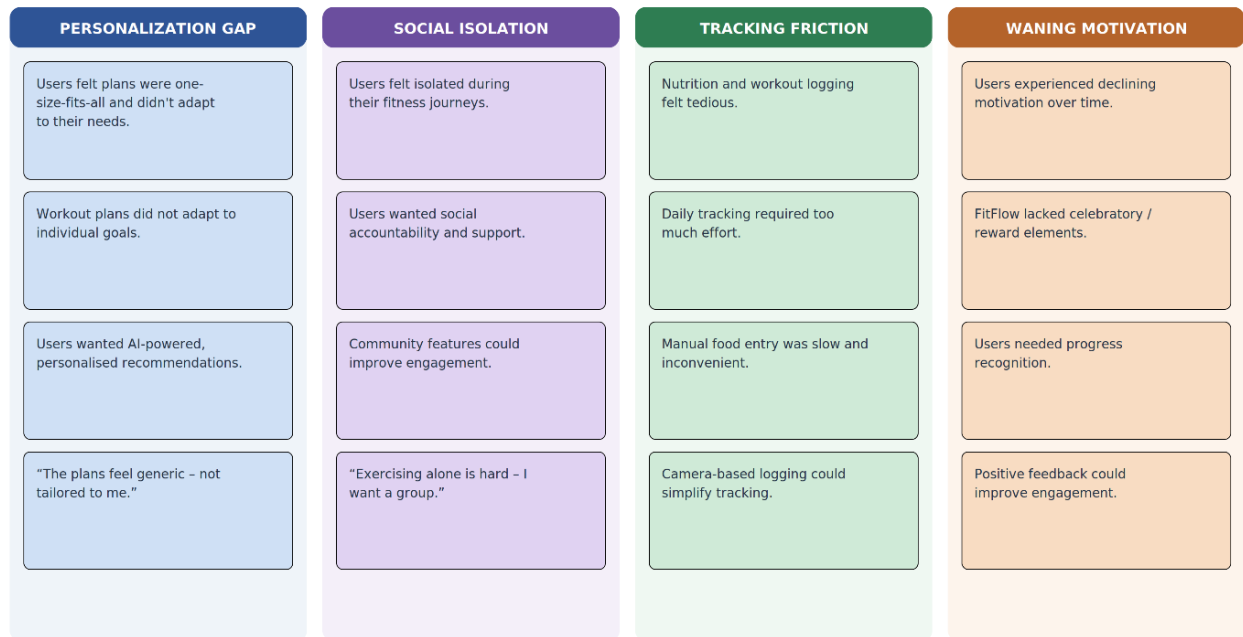
- Personalisation sits at the centre of the diagram – it connects to flexibility, onboarding, and motivation, confirming it as the top redesign priority.
- Motivation & Rewards and Social & Community overlap strongly: social features are the main lever for improving motivation and retention.
- Nutrition Tracking and Onboarding & Usability form a “friction” group – reducing effort in these areas directly addresses early drop-off.

Affinity Diagram (Miro Board)

The affinity board below groups the user-research findings into the four validated theme clusters.

FITFLOW USER RESEARCH - AFFINITY DIAGRAM

Grouping user-research findings into key themes



Important Quantitative Finding

68% of users abandoned FitFlow shortly after onboarding - an analytics signal (not one of the four validated themes) that warrants investigation into the onboarding experience.

Activity 4: Develop Three Personas and Empathy Maps

Three primary user segments were identified from the research data. A detailed persona and empathy map were created for each, supported by direct evidence from the Lab 01 findings.

Persona 1: Alex Rivera – Busy Professional (32)

Persona Card

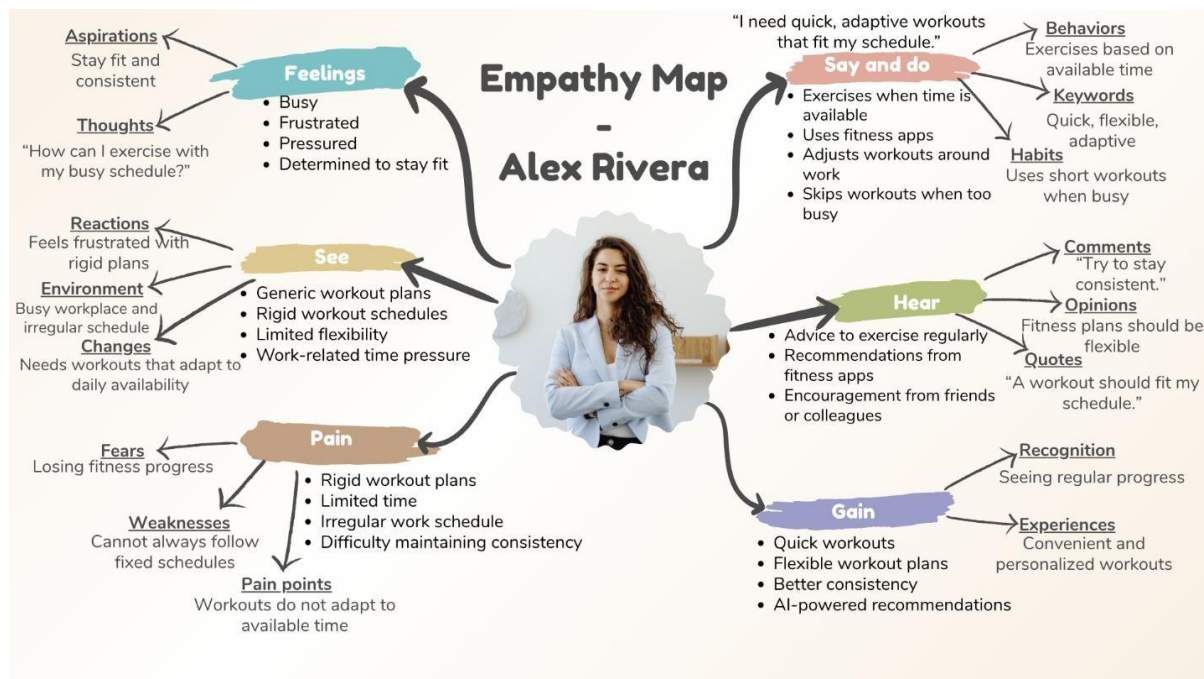
GOALS	• Maintain a consistent fitness routine • Exercise despite an irregular work schedule • Complete effective workouts in limited time	 ALEX RIVERA Age 32 Years Old Occupation Marketing Professional
NEEDS	• Quick workouts • Adaptive workout recommendations • Flexible fitness plans that fit changing schedules	
Frustrations	• Rigid workout plans • Plans that do not fit changing schedules • Difficulty maintaining consistency when work becomes busy	
"I need quick, adaptive workouts that fit my schedule."		

Persona Profile – Alex Rivera

Demographics	32, busy marketing professional, works long and irregular hours, uses fitness apps a few times a week.
Goals	Stay consistent despite a chaotic schedule; get quick, adaptive workouts that fit irregular time.
Behaviours	Opens the app in short bursts; abandons rigid fixed plans; relies on reminders.
Frustrations	Rigid plans don't adapt to his schedule; tracking progress feels like extra effort.

Empathy Map

Says	Thinks
"I want AI-generated adaptive workouts so I can stay consistent despite my chaotic schedule."	"Will this actually fit my week, or is it a waste of time?"
Does	Feels
Squeezes in short sessions; skips days when busy; ignores generic plans.	Frustrated and time-pressured; demotivated by rigid plans.
Pains	Gains
No flexibility; plans ignore his schedule; effort to log.	Quick adaptive AI workouts; smart reminders; easy progress feedback.



Persona 2: Priya Singh – Beginner Teacher (27)

Persona Card

GOALS	• Build a consistent fitness routine • Improve fitness safely and confidently • Stay motivated throughout her fitness journey	
NEEDS	• Safe social connections • Accountability partners • Support and encouragement • Clear and beginner-friendly fitness guidance	
Frustrations	• Feeling isolated during her fitness journey • Lack of accountability • Limited support and encouragement	

"I want safe social connections and someone to keep me accountable."

PRIYA SINGH

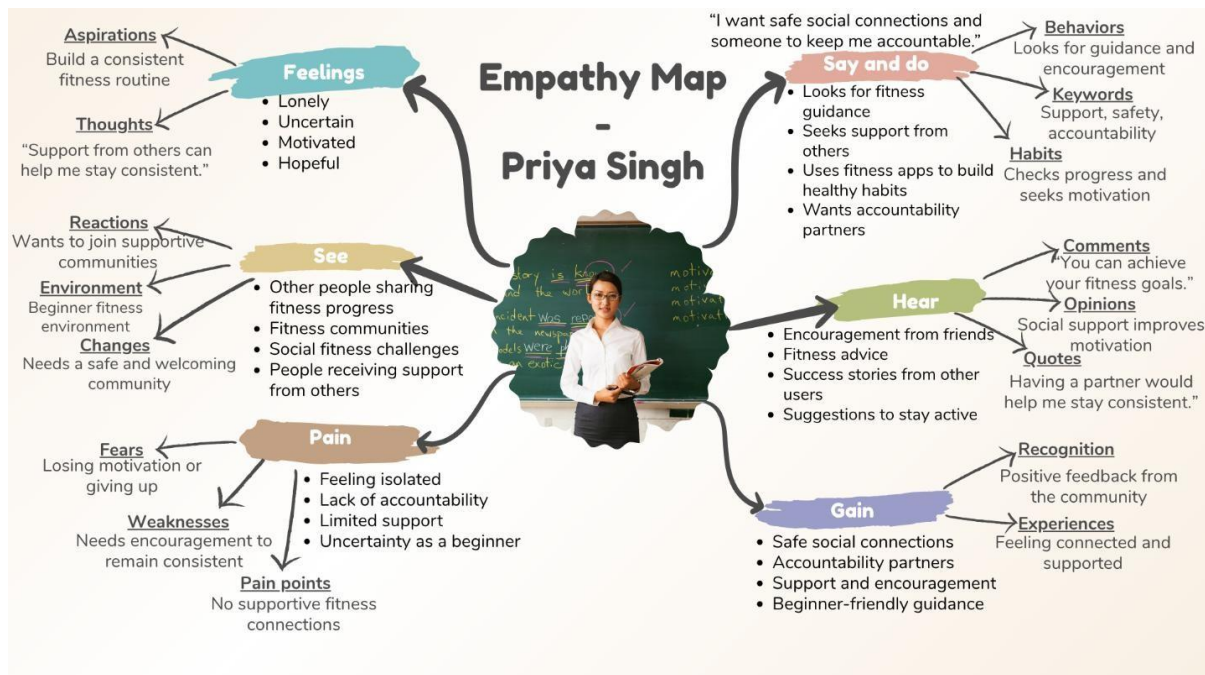
Age	27 Years Old
Occupation	Teacher
Fitness Level	Beginner

Persona Profile – Priya Singh

Demographics	27, beginner teacher, new to structured exercise, uses the app on most days but lacks confidence.
Goals	Build a safe, consistent routine; find safe social connections and accountability partners.
Behaviours	Follows guided sessions; looks for reassurance and community; motivated by encouragement.
Frustrations	Feels lonely during solo fitness journeys; unsure which exercises are safe; demotivated easily.

Empathy Map

Says	Thinks
"I want safe social connections and accountability partners."	"Would a supportive community make this less intimidating?"
Does	Feels
Starts guided workouts; hesitates during onboarding; seeks group support.	Nervous but hopeful; lonely during solo journeys; encouraged by small wins.
Pains	Gains
Loneliness; confusing onboarding; fear of doing exercises wrong.	Beginner-friendly AI coaching; safe supportive community; clear guidance.



Persona 3: Sam Fernando – Fitness Enthusiast (30)

Persona Card

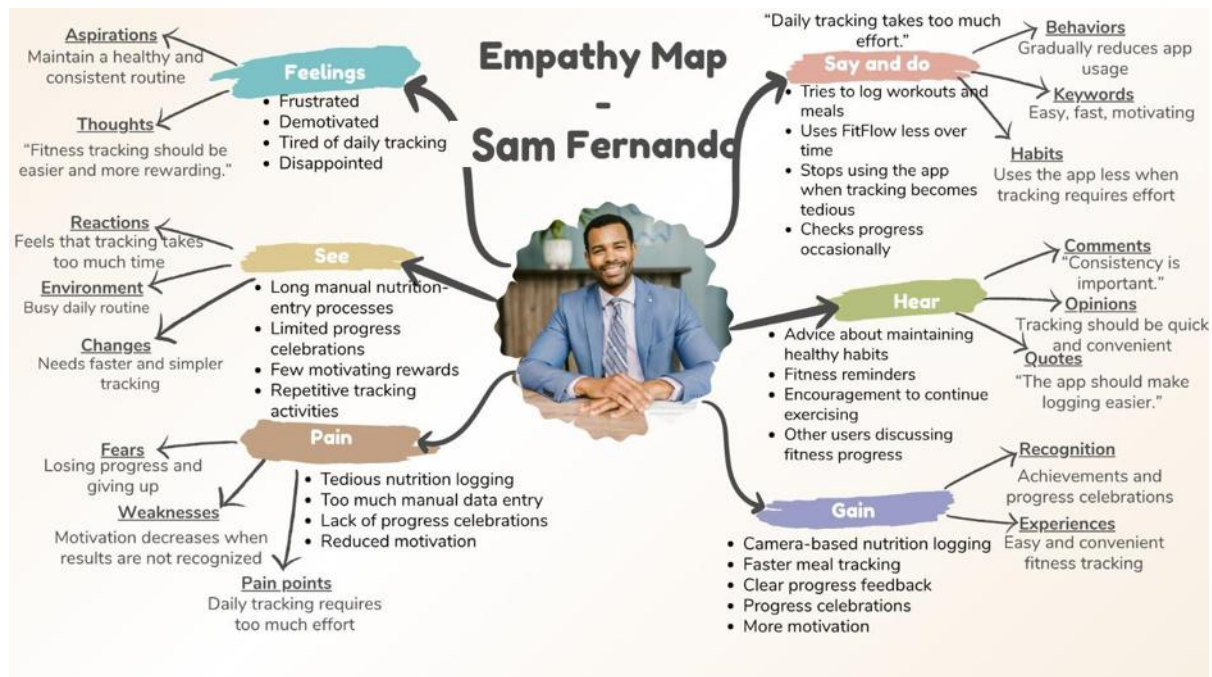
GOALS	• Maintain a consistent and healthy fitness routine • Stay motivated and continue working toward fitness goals • Monitor fitness progress easily	
NEEDS	• Easier workout and nutrition tracking • A faster method for logging meals • More motivation and encouragement • Clear progress feedback and achievements	
Frustrations	• Tedious nutrition logging • Lack of progress celebrations • Reduced motivation over time • Daily tracking requires too much effort	
"Daily tracking takes too much effort."		SAM FERNANDO Age 30 Years Old Occupation Office Administrator User Type Fitness Enthusiast

Persona Profile – Sam Fernando

Demographics	29, experienced gym-goer / part-time coach, tracks workouts and nutrition closely, an advanced user.
Goals	Optimise performance and nutrition; track detailed progress; keep training partners accountable.
Behaviours	Logs every session; reviews metrics; frustrated by tedious nutrition entry.
Frustrations	Manual food logging is slow; wants richer data, dashboards, and accountability tools.

Empathy Map

Says	Thinks
"Manual food logging takes too long – I stop after a few days."	"Can the app give me deeper data and keep others accountable?"
Does	Feels
Tracks detailed metrics; abandons slow logging; supports training partners.	Motivated but impatient with friction.
Pains	Gains
Tedious nutrition logging; limited advanced insights.	Camera-based food logging; detailed progress dashboards; group accountability.



Activity 5: Write User Stories

Using insights from the personas, themes, and affinity diagram, the following user stories were written in the standard format and prioritised by user impact and business value. Priority is marked High (H), Medium (M), or Low (L).

#	User Story	Priority
US1	As a busy professional, I want AI-generated adaptive workout plans so that I can stay consistent despite my chaotic schedule.	High
US2	As a beginner, I want AI coaching that suggests safe, suitable exercises so that I feel confident starting out.	High
US3	As a beginner, I want a simple, guided onboarding so that I reach my first workout without getting lost.	High
US4	As a user, I want personalised workout plans based on my goals so that the app feels tailored to me.	High
US5	As a fitness enthusiast, I want camera-based food logging so that tracking nutrition is quick and I keep doing it.	High
US6	As a user, I want to join challenges with friends so that I stay motivated and accountable.	Medium
US7	As a beginner, I want to join supportive community groups so that I don't feel alone while exercising.	Medium
US8	As a user, I want achievement badges and streaks so that I feel rewarded for staying consistent.	Medium
US9	As a busy user, I want smart reminders so that I don't forget to work out on hectic days.	Medium
US10	As a fitness enthusiast, I want a detailed progress dashboard so that I can review my performance over time.	Medium
US11	As a coach, I want to support clients in a group so that I can keep them accountable.	Low
US12	As a user, I want to share achievements on my feed so that I can celebrate progress with others.	Low

Activity 6: Create User Journey Maps

A current-state journey and a future/ideal-state journey were mapped for the Busy Professional persona (Alex), from initial awareness to long-term usage, referencing specific FitFlow features such as workout logging, nutrition tracking, and social elements.

Current-State Journey (Alex)

Stage	Actions	Thoughts / Emotions	Pain Points
Awareness	Sees FitFlow in the app store; reads mixed reviews.	Curious but cautious – “will it fit my schedule?”	Rating dropped to 3.8; reviews mention generic plans.
Onboarding	Signs up; sets up profile.	Impatient; wants to start quickly.	Setup feels long; unclear next step.
First use	Opens a workout plan; tries to log a session.	Disappointed – plan feels generic.	Plan doesn’t adapt to his time; logging is fiddly.
Ongoing use	Tries nutrition logging; skips days.	Frustrated; loses momentum.	Manual food logging is tedious; no motivation hooks.
Long-term	Opens the app less; considers quitting.	Indifferent – “not worth it.”	Churn risk – part of the 68% drop-off.

Future / Ideal-State Journey (Alex)

Stage	Actions	Thoughts / Emotions	Opportunities / Solutions
Awareness	Sees improved reviews highlighting AI workouts.	Interested – “this adapts to me.”	Market AI personalisation and flexibility.
Onboarding	Completes a short guided setup.	Reassured; reaches first workout fast.	Streamlined onboarding reduces drop-off.
First use	Gets an AI workout sized to his free time.	Impressed – “finally fits my week.”	AI adaptive workouts; one-tap logging.

Stage	Actions	Thoughts / Emotions	Opportunities / Solutions
Ongoing use	Uses camera-based food logging; joins a challenge.	Motivated; feels accountable.	Camera nutrition logging; social challenges; badges.
Long-term	Keeps streaks; reviews progress dashboard.	Loyal – “this keeps me on track.”	Retention up; rating recovers toward 4.5.

The emotional arc moves from cautious and frustrated (current state) to reassured, motivated, and loyal (future state) as the redesign removes friction and adds personalisation and social motivation.

Visual Journey Maps



Journey Map – Alex Rivera (Busy Marketing Professional)

USER JOURNEY MAP — PRIYA SINGH

Key scenario: Building a consistent routine as a beginner

User: Priya Singh — Beginner Teacher (Primary User)

	DISCOVER	ONBOARD	FIRST WORKOUT	ONGOING USE	LONG-TERM
DOING	- Hears about FitFlow - from a friend	- Signs up - Sets beginner goals	Starts a guided workout	- Joins a group - Tracks progress	Builds a routine
THINKING	"Is this for beginners?"	"Where do I start?"	"Am I doing it safely?"	"Do others support me?"	"Can I keep this up?"
TOUCHPOINTS	Friend / social media	Onboarding & goals	Guided workout screen	Community & challenges	Progress & rewards
WEAKNESSES	Unclear if beginner-friendly	Overwhelming exercise library	Fear of wrong form	Feels isolated alone	Motivation drops
OPPORTUNITIES	Beginner onboarding path	AI coaching for beginners	Safe exercise guidance	Supportive community groups	Encouragement & badges

User Journey Map – Priya Singh (Beginner Teacher)

USER JOURNEY MAP — RYAN FERNANDO

Key scenario: Re-engaging a lapsed user with easier tracking

User: Ryan Fernando — Office Administrator, Lapsed FitFlow User (Primary User)

	RETURN	RE-ONBOARD	LOG ACTIVITY	ONGOING USE	LONG-TERM
DOING	Re-opens FitFlow	Updates profile & goals	Tries logging meals	- Uses camera logging - Sees streaks	Stays engaged
THINKING	"Is it easier now?"	"Quick to restart?"	"Is logging faster?"	"Am I being rewarded?"	"Worth staying?"
TOUCHPOINTS	Re-engagement notification	Re-onboarding flow	Nutrition logging screen	Camera logging & streaks	Progress celebrations
WEAKNESSES	Remembers old frustrations	Repeat setup effort	Manual entry still slow	No celebrations before	Motivation had faded
OPPORTUNITIES	Win-back messaging	Fast re-onboarding	Camera-based logging	Streaks & rewards	Progress celebrations

User Journey Map – Sam Fernando (Fitness Enthusiast)

Activity 7: Generate How Might We Statements

Key pain points and opportunities from the previous activities were reframed into optimistic, open-ended “How Might We” (HMW) questions that inspire creative solutions. The statements focus on the three redesign priorities.

Focus Area	Pain Point	How Might We Statement
Personalisation	Plans feel generic and users don't trust the recommendations.	How might we make personalization feel truly intelligent and trustworthy?
Social engagement	Users feel isolated but can be overwhelmed by social features.	How might we foster positive social connections without overwhelming users?
Nutrition tracking	Manual food logging is tedious and abandoned.	How might we make nutrition tracking so quick and easy that users keep doing it?
Onboarding (extra)	68% of new users drop off after onboarding.	How might we get new users to their first successful workout as quickly as possible?

These HMW statements were prioritised with the team, with personalisation, social engagement, and nutrition tracking taken forward as the primary drivers of the redesign.

Activity 8: Functional & Non-Functional Requirements with Traceability

Insights from all previous activities were translated into clear functional and non-functional requirements supporting the new features (AI personalised plans, social community, nutrition tracking) and the overall redesign. Each requirement is prioritised High / Medium / Low.

Functional Requirements

ID	Functional Requirement	Priority	Source
FR1	The system shall provide an AI workout engine that generates adaptive, personalised plans from user goals and schedule.	High	US1, US4
FR2	The system shall offer beginner-friendly AI coaching with safe exercise suggestions.	High	US2
FR3	The system shall provide a short, guided onboarding flow leading to a first workout.	High	US3
FR4	The system shall support camera-based nutrition logging powered by computer vision.	High	US5
FR5	The system shall provide private social circles with robust privacy controls.	Medium	US6, US7
FR6	The system shall let users create and join challenges and group interactions.	Medium	US6
FR7	The system shall award badges and track streaks for consistency.	Medium	US8
FR8	The system shall send smart, personalised workout reminders.	Medium	US9
FR9	The system shall present engaging progress dashboards.	Medium	US10

Non-Functional Requirements

ID	Non-Functional Requirement	Priority	Source
NFR1	The app shall deliver fast performance, generating AI suggestions within a few seconds.	High	Theme: Flexibility

ID	Non-Functional Requirement	Priority	Source
NFR2	The app shall provide offline capabilities so core tracking works without a connection.	High	Case study
NFR3	The app shall meet accessibility compliance standards and be usable for beginners.	High	Persona: Priya
NFR4	The app shall enforce strong data security, compliant with GDPR and CCPA.	High	Lab 01 ethics
NFR5	Onboarding shall be completable quickly to reduce the 68% drop-off.	Medium	Theme: Onboarding

Traceability Matrix

The traceability matrix links each requirement back to the themes, personas, user stories, and HMW statements it originated from, ensuring every requirement is grounded in the research.

Req.	Theme	Persona	User Story	HMW
FR1	Personalisation / Flexibility	Alex	US1, US4	HMW-Personalisation
FR2	Personalisation	Priya	US2	HMW-Personalisation
FR3	Onboarding friction	Priya	US3	HMW-Onboarding
FR4	Nutrition tracking	Sam	US5	HMW-Nutrition
FR5	Motivation / Social	Alex, Priya	US6	HMW-Social
FR6	Social isolation	Priya	US7	HMW-Social
FR7	Poor motivation	Alex	US8	HMW-Social
FR9	Personalisation	Sam	US10	HMW-Personalisation

The requirements were reviewed for clarity, feasibility, and completeness, confirming that each redesign priority is fully traceable from the original research through to an actionable requirement.